SST control architecture (overview)

monolithic control layout
high bandwidth PI and RPI

distributed controller based on master/slave

distributed control layout

fundamental isolated DC/AC

phase cluster
set of N fundamental isolated DC/AC

whole three phase SST
base on phase clusters

single-phase-inverter current control and DC link voltage control by DAB phase shift

distributed controller based on master/slave architecture
current control: master
voltage control: slaves

global distributed controller based on cost function for power flow regulation

$$\vec{u}_g^{ac}(t) = \begin{bmatrix} u_u(t) & u_v(t) & u_w(t) \end{bmatrix}^T$$
$$\vec{i}_g^{ac}(t) = \begin{bmatrix} i_u(t) & i_v(t) & i_w(t) \end{bmatrix}^T$$
$$u_g^{dc}(t)$$

$$\hat{\vartheta}_{pll}(t) \quad \vec{u}_{\xi\eta}^{p}(t) \quad \vec{u}_{\xi\eta}^{n}(t) \quad \vec{i}_{\xi\eta}^{p}(t) \quad \vec{i}_{\xi\eta}^{n}(t) \quad u_g^{dc}(t)$$